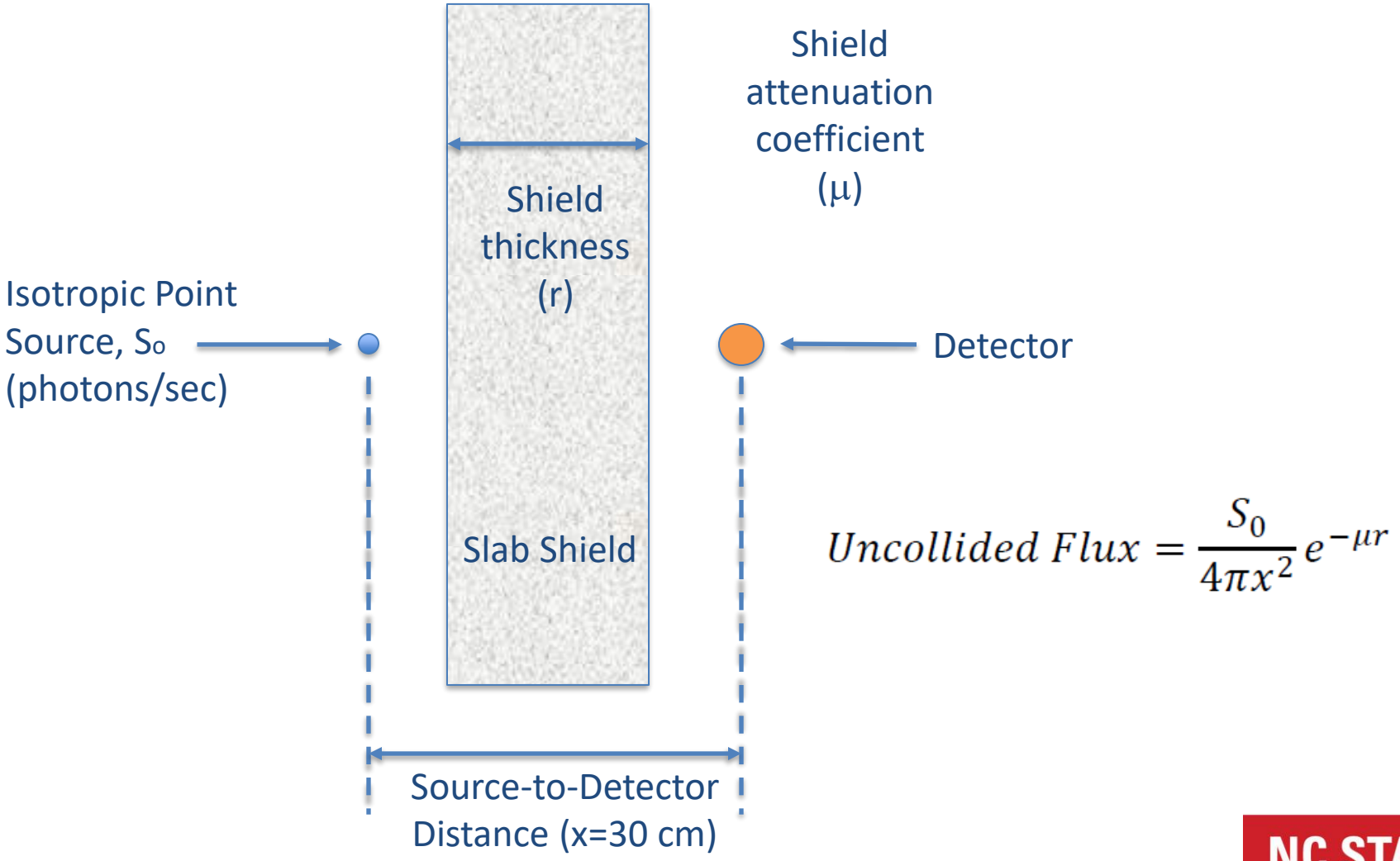
Photon Buildup Factor Problem

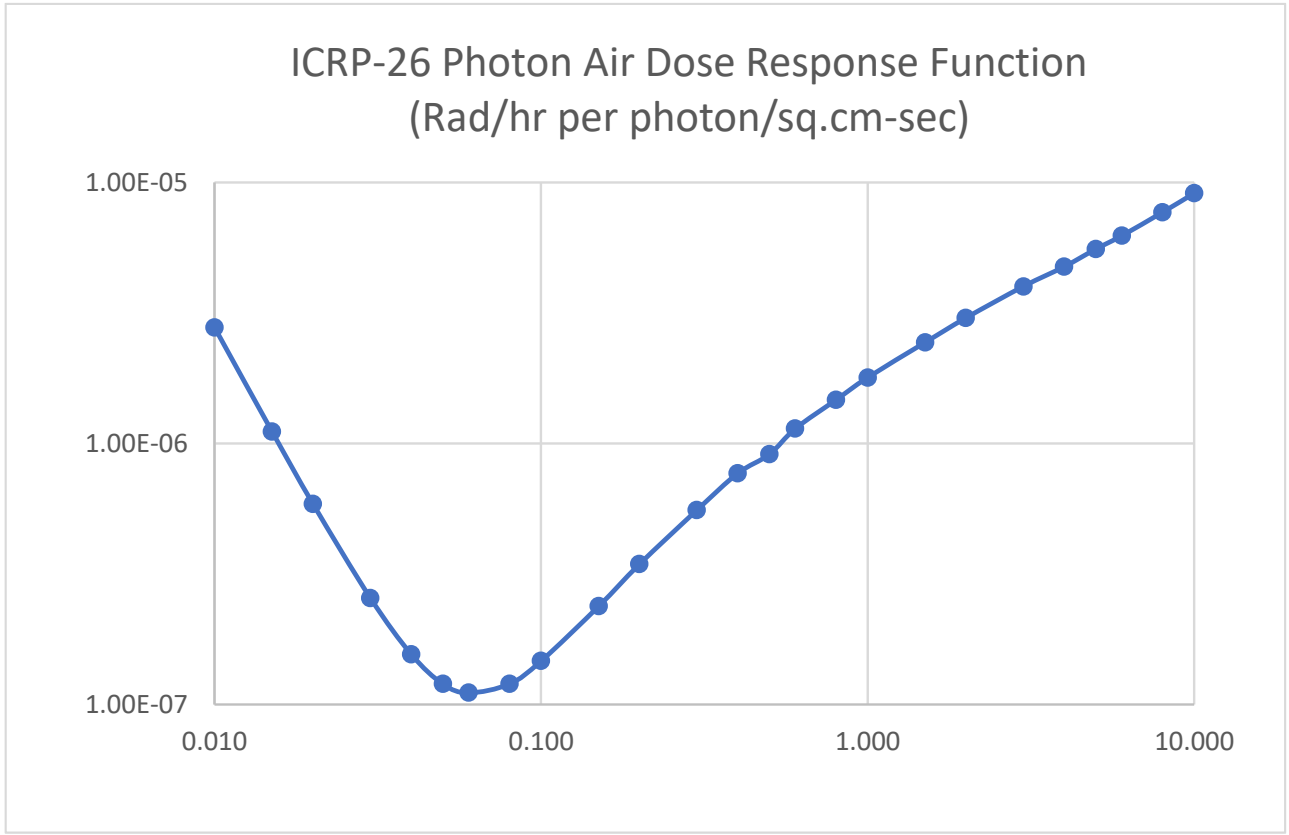


$$Uncollided\ Flux = \frac{S_0}{4\pi x^2} e^{-\mu r}$$

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Photon Buildup Factors

$$Uncollided\ Dose\ Rate = DF(E)\frac{S_0}{4\pi x^2}e^{-\mu r}$$



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Photon Buildup Factors

Goldstein-Wilkins Build-Up Factors

Lead			Buildup Factors			
				μr		
E (Mev)	μ(cm-1)	1	2	4	7	10
0.5	1.746	1.24	1.42	1.69	2.00	2.27
1.0	0.782	1.37	1.69	2.26	3.02	3.74
2.0	0.510	1.39	1.76	2.51	3.66	4.84
3.0	0.472	1.34	1.68	2.43	3.75	5.30
4.0	0.469	1.27	1.56	2.25	3.61	5.44
6.0	0.492	1.18	1.40	1.97	3.34	5.69
8.0	0.521	1.14	1.30	1.74	2.89	5.07
10.0	0.549	1.11	1.23	1.58	2.52	4.34

$$Total\ Dose\ Rate = DF(E)B(\mu, r)\frac{S_0}{4\pi x^2}e^{-\mu r}$$

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MCNP Calculation of BU Factors

Part 1:

- Assume source is 1.0e+7 photons per sec
- Consider fixed source-to-detector distance (x = 30 cm center-to-center)
- Consider isotropic point source with the following incident photon energies:
  - 0.5, 1.0, 2.0, 3.0, 4.0, 6.0, 8.0 and 10.0 MeV
- Simulate air and tissue "equivalent" photon detectors using given photon air and tissue dose response functions provided
- Consider slab shields of Lead
- Consider various shield (μ,r) values of 1, 2 ,4, 7 and 10
- Calculate B(μ,r) for all photon energies, shield thicknesses, and shield materials by ratioing the total to uncollided dose rate responses
- Compare MCNP calculated build-up factors to the Goldstein-Wilkins build-up values provided

$$B(\mu, r) = \frac{DR_{Total}}{DR_{Uncollided}} = \frac{DR_{Uncollided} + DR_{Collided}}{DR_{Uncollided}} = 1 + \frac{DR_{Collided}}{DR_{Uncollided}}$$

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MCNP Calculation of BU Factors

Part 2:

- Assume same source and source-to-detector distance as Part 1
- Consider isotropic point source with the same incident photon energies
- Consider slab shields of Lead
- Consider various shield (μ,r) values of 1, 2 ,4, 7 and 10
- Calculate MCNP B(μ,r) factors for all photon energies, shield thicknesses, and shield materials by ratioing total to un-collided dose rate responses
- Calculate Taylor build-up factors using equations and data provided
- Compare MCNP calculated B(μ,r) values to Taylor B(μ,r) factors

Taylor Form of Build-up Factor Equation:

$$B(\mu, r) = A_1 e^{-\alpha_1 \mu r} + A_2 e^{-\alpha_2 \mu r}$$

Where:

$$A_2 = 1 - A_1$$

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Taylor Coefficients

Lead			
Taylor Buildup Factor Coefficients			
E (Mev)	A_1	α_1	α_2
0.5	1.677	-0.03084	0.30941
1.0	2.984	-0.03503	0.13486
2.0	5.421	-0.03482	0.04379
3.0	5.580	-0.05422	0.00611
4.0	3.897	-0.08468	-0.02383
6.0	0.926	-0.17860	-0.04635
8.0	0.368	-0.23691	-0.05684
10.0	0.311	-0.24024	-0.02783

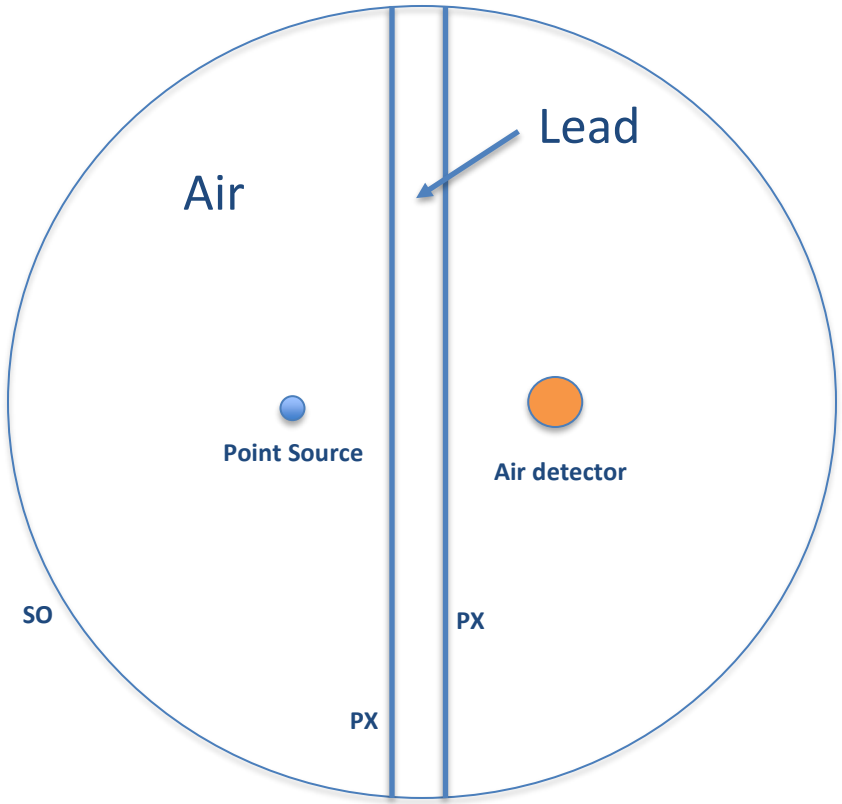
$$B(\mu, r) = A_1 e^{-\alpha_1 \mu r} + A_2 e^{-\alpha_2 \mu r}$$

Where:

$$A_2 = 1 - A_1$$

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Example Input File Model




```
      Photon Build-Up Factor Problem
1     1 0.0000538      -1 -2     imp:p=1
2     2 0.032965307    -1  2 -3  imp:p=1
3     1 0.0000538      -1  3 -4  imp:p=1  $ Detector
4     1 0.0000538      -1  3  4  imp:p=1
5     0                 1        imp:p=0

1     so  500.0
2     px  0.0
3     px  10.0
4     sx  25.0 2.0

mode  p
nps   1.0e+08
sdef  erg=10.0  x=-5.0  y=0.0  z=0.0  par=2
f14:p 3    $ ICRP-26 Photon Air Dose Response Function (Rad/hr/cm2-s)
de14  0.01 0.015 0.02 0.03 0.04 0.05 0.06 0.08 0.1 0.15 0.2 0.3 0.4
      0.5 0.6 0.8 1.0 1.5 2.0 3.0 4.0 5.0 6.0 8.0 10.0
df14  2.78e-6 1.11e-6 5.88e-7 2.56e-7 1.56e-7 1.2e-7 1.11e-7
      1.2e-7 1.47e-7 2.38e-7 3.45e-7 5.56e-7 7.69e-7 9.09e-7
      1.14e-6 1.47e-6 1.79e-6 2.44e-6 3.03e-6 4.0e-6 4.76e-6
      5.56e-6 6.25e-6 7.69e-6 9.09e-6
e14   0.01 9.999 10.0
f24:p 3    $ DS02 Photon Tissue Dose Response Function (Rad/hr/cm2-s)
de24  0.01 0.015 0.02 0.03 0.04 0.05 0.06 0.08 0.1 0.15 0.2 0.3 0.4
      0.5 0.6 0.8 1.0 1.25 1.5 2.0 3.0 4.0 5.0 6.0 8.0 10.0 15.0 20.0
df24  2.61e-06 1.09e-06 5.90e-07 2.53e-07 1.52e-07 1.17e-07 1.07e-07
      1.18e-07 1.45e-07 2.38e-07 3.39e-07 5.47e-07 7.52e-07 9.43e-07
      1.13e-06 1.47e-06 1.77e-06 2.12e-06 2.43e-06 2.98e-06 3.92e-06
      4.72e-06 5.47e-06 6.16e-06 7.52e-06 8.89e-06 1.22e-05 1.56e-05
e24   0.01 9.999 10.0
m1    7014    0.0000425     $ Air at STP
      8016    0.0000113
m2    82000   0.032965307   $ Lead Shield
```

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Things to Consider …

- You will likely need to run approx. 1.0e+08 particle histories for the lower energy (< 4 MeV) and thin shield ($\mu r < 4$) cases to get reasonable statistics on the detector tally
- Consider running 1.0e+09 to 1.0e+10 particle histories for higher energy and thicker shield cases since sampling of the detector will be lower
- Use of the F15 point detector tally for higher energy and thicker shield cases can help improve statistical uncertainty of detector response
- Goldstein-Wilkins and Taylor build-up factors should agree reasonably well with MCNP results for lower energy and thin shield cases but will likely not agree very well for higher energy and thick shield cases
- Look carefully at the table of photon production and loss terms in the MCNP output file to try and understand cases where they may not agree and give an explanation of your results

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